

Research paper

Analysis on influence factors of static and dynamic response of prefabricated prestressed steel-concrete hybrid tower for onshore wind turbines

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ABSTRACT

To clarify the effect of damage on the dynamic response of the wind tower and verify the FE method, shaking table tests of the simplified model are carried out. The results reveal that the acceleration response at the tower top exhibits a distinct beat vibration phenomenon under specific external load excitations. Subsequently, a series of finite element analyses is carried out to study the static and dynamic characteristics of the prefabricated prestressed steel-concrete hybrid tower. Under static load, variations in structural stiffness and joint configurations induce stress redistribution, significantly affecting the global stress state of the tower. Compared to the integral tower, the presence of horizontal joints has a negligible impact on the mode shape and natural frequency of the wind turbine tower, with the maximum reduction in each frequency order being only 0.79 %. The appearance of vertical joints caused a significant decrease in the frequencies of each order of the tower, with the reduction in the frequencies of each order of modes reaching 4.94 %, 24.64 %, and 16.58 %, respectively. When the joints of the concrete tower section are damaged to varying degrees, the impact on the stress and deformation of the prestressed hybrid tower is almost identical for different joint designs. Furthermore, the impact of joint damage on the frequencies of each order in the hybrid tower with different joint conditions is also almost uniform. The research findings provide important technical support for the design and fatigue risk prevention of hybrid tower structures.

1. Introduction

With the continuous increase in global energy demand, the application of renewable energy has become a trend, and the market for wind power is constantly expanding [1–7]. As wind turbine capacity continues to increase, the hub height of wind turbines is also continuously being raised in order to achieve higher energy capture efficiency. Traditional all-steel towers face challenges such as transportation limitations, heavy weight, and buckling sensitivity once their height exceeds 120 m. Concrete towers, on the other hand, suffer from drawbacks such as excessive mass and long construction periods, making them unsuitable for the high tower requirements of large-megawatt onshore wind turbines [8,9]. Prestressed steel-concrete hybrid towers, composed of a

high-rigidity concrete lower tower section and a lighter steel lower tower section, combine ease of manufacturing with high rigidity, making them an effective solution for increasing tower height [9–13]. Their share in the onshore wind power market has risen rapidly in the past five years, becoming the mainstream engineering solution for high towers (>150 m) [14–18]. However, compared to traditional single-material steel towers and concrete towers, prefabricated prestressed steel-concrete hybrid tower structures are more complex in terms of dynamic characteristics [19,20]. The unevenness of materials used in the tower body, the large number of horizontal and vertical joints caused by the segmented prefabrication and assembly method, and the overall structural stiffness degradation during service can all affect the natural frequency of the structure, and thus affect the vibration damping effect

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with the excitation frequencies of the wind turbine 1P, 3P, and 6P

The stepped variable diameter tower technology originated from the independent development of Spain, the United States, and China. Its structural feature is the use of tower sections with different diameters that are "larger at the bottom and smaller at the top", which can give full play to the structural strength. Mechanically, it functions as a tapered cantilever column. Based on the different joint splicing methods of the concrete sections, prefabricated prestressed steel-concrete hybrid towers can be divided into two types: segmented concrete towers and piece-type concrete towers. Among them, the segmented hybrid tower structure divides the concrete tower section into multiple circular segments along its height. The entire tower has no vertical joints, and the horizontal joints between the tower segments are connected by grouting material or epoxy resin structural adhesive. The segmented hybrid tower structure is designed for easy transportation. Based on the segmented hybrid tower, each hybrid tower segment is further divided into 2 to 4 concrete segments along the height direction. During assembly, the concrete segments of each segment are first assembled into a ring, and then the vertical joints between the hybrid tower segments are grouted to connect them. After the segments have gained strength, they are then vertically hoisted.

However, under external loads, the top of the tower will experience large displacement, and the weight of components such as blades and hubs at the top of the tower will bring about a very obvious second-order gravitational effect (P- Δ effect). Meanwhile, different jointing schemes for the hybrid tower will result in significant geometric nonlinearity in the structure, which also brings certain difficulties to the structural stress analysis. N.Bazeos et al. [21] conducted a static analysis on a 450KW wind turbine tower. Using the finite element method (FEM), they studied the influence of the doorway and the arrangement of stiffening ribs around it on the overall structural performance of the tower, focusing on the shear stress on both sides of the doorway. Furthermore, it accurately and effectively determined the minimum load combination that would cause local buckling in the shell structure. The calculation of this load combination is directly related to many geometric variables related to the arrangement of the structural load-bearing components. De Lana et al. [8] studied the static performance of a 100 m high prestressed concrete wind turbine tower, including its deformation and stress distribution. Finally, they achieved structural optimization through a genetic algorithm, providing experience for the optimization design of the tower.

Current research on prestressed hybrid tower structures mainly focuses on the load-bearing characteristics of joints under static loads, hybrid tower construction methods, and overall structural optimization [22,23]. Research on the dynamic characteristics of the structure, especially higher-order vibration modes, is relatively limited. The dynamic characteristics analysis of tall towers is very important. The first two frequencies of the tower should be avoided from the harmonics (1P, 3P, 6P) of the wind turbine to avoid resonance and fatigue problems [24–27]. Quilligan A et al. [28] studied the dynamic performance of wind turbine towers. Using FE simulation, they modeled and analyzed steel towers and prestressed concrete towers, respectively, compared the dynamic performance of the two structures, and pointed out that the prestressed concrete tower has better dynamic performance when the tower height is higher. P.J.Murtagh et al. [29] used an analytical method to approximate the tower as a simplified cantilever beam with a fixed concentrated load at the top, and obtained the first few natural frequencies of the tower. They compared the results of the analytical method with the results of the FEM, and the two results were in good agreement. This verifies that the model reduction technique can obtain the accurate modal characteristics of the first three natural frequencies while reducing the complexity of the tower model, and this simplification method is feasible. Lee and Bang [30] conducted a numerical analysis on a tower that collapsed on Jeju Island, South Korea. They performed linear and nonlinear buckling analyses on the tower to predict the critical buckling load of the tower. Through simulation analysis,

they finally found the critical load range of the tower collapse. Paredes and Barbat [31] conducted a numerical simulation study based on an actual four-segment reinforced concrete tower structure. They performed FE simulations on different concrete materials and concrete constitutive structures. The results showed that the presence of the connection nodes between concrete segments of the concrete tower does not affect the overall structural stiffness, but will significantly affect the area of overall structural damage. At the same time, they developed a calculation program for wind turbine concrete towers under gap cyclic load. This program provides a reference for the dynamic analysis of such structures.

Currently, simplified analysis methods are generally used for the overall stress analysis of towers in actual engineering projects, and the impact of different jointing schemes on the overall stress of hybrid tower structures is still unclear. In addition, at present, the first-order fundamental frequency resonance problem of towers is given more attention in the design and monitoring stages, but less attention is paid to higher-order resonance and the fatigue problems it causes. The amplitude of the hybrid tower is affected by factors such as local wind conditions, vibrations generated during the operation of the wind turbine, and dynamic parameters of the hybrid tower. Although the amplitude of higher-order resonance is smaller than that of first-order resonance, it causes greater fatigue damage to the tower and will quickly consume the tower's lifespan. Therefore, further in-depth research is urgently needed on the dynamic response of the hybrid tower structure and its influencing factors.

To address the knowledge gap regarding the effects of joint configurations and structural damage on the static and dynamic responses of prefabricated prestressed hybrid towers, this study employs a combination of indoor model tests and numerical simulation analysis. First, indoor shaking table model tests are conducted to verify the correctness of the FEM and analyze the dynamic response of the structure under damage conditions. Next, a high-precision FE model of the prestressed hybrid tower was established, and a series of parametric analyses was conducted on the effects of joint schemes on the overall structural stress, deformation, frequencies, and mode shapes of the tower. Furthermore, the effects of different degrees of seam damage on the static and dynamic response of the hybrid tower structure are compared and analyzed, providing theoretical support for the further promotion and application of the ultra-high prestressed hybrid tower structure in the wind power market.

2. Shaking table test

In order to verify the accuracy of the numerical simulation method and explore the real dynamic response of high-rise tower structures under damage conditions, shaking table tests were conducted and corresponding FE simulations were established for comparison and analysis. The small-scale shaking table test design of prestressed hybrid tower structure model is a complex problem, which involves synchronous scaling of different materials and additional prestressed systems for steel tower sections and concrete tower sections, making it difficult to accurately set structural damage. Meanwhile, the cost of the model is high, the production process is complex, and the quality of the model is difficult to guarantee. In order to achieve the experimental research objectives under the premise of meeting the minimum requirements for model processing and experimental conditions [32], a 1/20 scale shaking table test was conducted using a 2MW wind power generation steel tower from a certain actual project as the prototype.

2.1. Test setup

A shaking table model test with a scale ratio of $S_l=1/20$ was conducted using a 2MW wind power tower as a prototype. The model was made of Q345 steel with an elastic modulus and stress similarity ratio $S_\sigma = S_E = 1$. Based on the additional mass and installation conditions,

Table 1

The similarity relationship and similarity coefficient of the test model.

Parameters	Similarity relationship	Similarity coefficient
Length l	S_l	1/20
Elasticity modulus E	S_E	1
Gravitational acceleration g	S_g	1
Acceleration a	S_a	2
Stress σ	S_σ	1
Strain ε	$S_\varepsilon = S_\sigma S_\varepsilon^{-1}$	1
Frequency f	$S_f = S_l^{-0.5} S_a^{0.5}$	6.3246
Period T	$S_T = S_l^{0.5} S_a^{-0.5}$	0.1581

the acceleration similarity ratio $S_a=2$ was determined. The physical parameters and similarity ratios are shown in Table 1. The tower structure is divided into 3 sections, each with a height of 1 m. The thicknesses of the three sections are 4 mm, 3 mm and 3 mm, respectively. The diameter of the tower tip is 200 mm and the diameter of the tower top is 148 mm. The diameter decreases linearly along the height of the tower. The detailed dimensions of the tower are shown in Table 2. The cabin of the tower is 600 mm in length, 300 mm in width, 230 mm in height, 20 mm in thickness, and 75 kg in weight. The center of gravity is located 50 mm behind the tower. The blade is simplified as a rectangular section, with a length of 1000 mm, a width of 30 mm, a thickness of 10 mm, and a mass of 6.5 kg. The hub is approximately replaced by a cylinder with a diameter of 200 mm and a height of 120 mm. The weight of the hub is 30 kg, and the center of gravity is 125 mm in front of the tower. The overall layout of this test is shown in Fig. 1.

The tower section is connected with the tower section by a flange with a thickness of 10 mm, and the two tower flange plates are connected with high-strength bolt M14. The connection between the tower section and the flange plate is welded, and the bottom tower is perforated and anchored on the shaking table. The additional mass of the tower is evenly distributed along the height and bound on the tower drum with an iron wire. In order to prevent the collision between the additional mass block and the tower body during the loading process, a rubber pad is placed between the mass block and the tower body.

0.2 g EI Centro wave was selected for unidirectional loading. Acceleration sensors are arranged along the loading direction at the heights of 0 m, 1 m, 2 m and 3 m, respectively. In the shaking table test, the tower

section is connected by 12 evenly distributed bolts at the connection flange, as shown in Fig. 1. To simulate structural damage, specifically the decrease in local stiffness and section deterioration between tower segments, a macro-equivalent damage simulation method was adopted by completely removing three bolts along the loading direction from the connecting flange. This method of bolt removal is widely verified in benchmark dynamic tests, as a reliable paradigm to idealize the local concrete cracking and resulting stiffness degradation typically observed in full-scale prestressed hybrid wind turbine towers [33]. Accurately replicating microscopic concrete tensile cracking in a scaled dynamic model is highly challenging due to significant size effects. The physical rationale behind this idealization is to replicate the nonlinear "breathing crack" mechanism. As demonstrated in previous studies [34], the core dynamic signature of a concrete crack is its opening under tension (loss of stiffness) and closing under compression (restoration of stiffness), which mathematically acts as a bilinear equivalent hinge [35]. By removing the tension-side bolts, the model flange permits exact interface

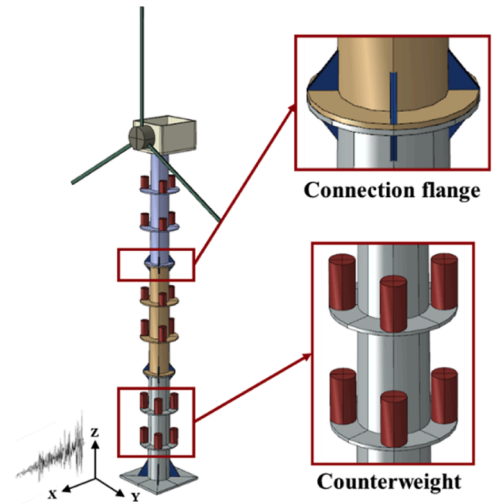


Fig. 2. FE model of the model tower.

Table 2

Parameters of the model.

Tower Section	Height (m)	Diameter (mm)	Tower wall thickness (mm)	Tower weight (kg)	counterweight weight (kg)	Total weight (kg)
Tower Section 1	1	Φ200~Φ183	4	19	32	51
Tower Section 2	1	Φ182~Φ165	3	13	21	34
Tower Section 3	1	Φ165~Φ148	3	12	18	30

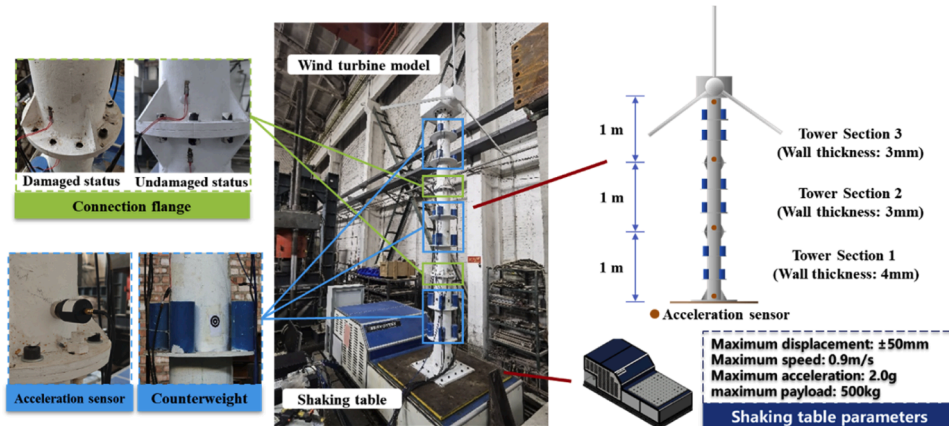


Fig. 1. Shaking table test model.

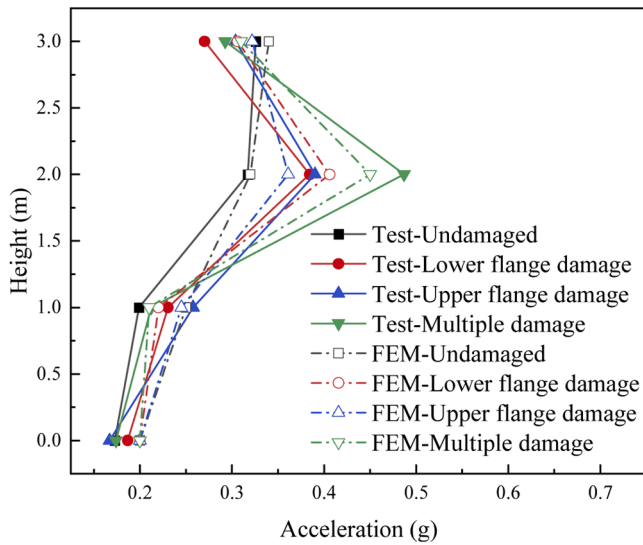


Fig. 3. Acceleration envelope of tower under different damage conditions.

separation and compression-contact, physically replicating this asymmetric stiffness degradation.

Based on this methodology, single structural damage is simulated by completely removing three bolts along the loading direction from a single connecting flange, while the multiple damage condition is simulated by removing three bolts from each connecting flange. Specifically, four test scenarios were carried out in this test: the undamaged condition, the single damage condition at the lower connecting flange (between tower Section 1 and tower Section 2), the single damage condition at the upper connecting flange (between tower Section 2 and tower Section 3), and the multiple damage condition involving both the upper and lower connecting flanges.

2.2. Numerical model

The FE model corresponding to the shaking table test was established via ABAQUS software, as illustrated in Fig. 2. Geometric dimensions of the model were consistent with the physical specimen. Steel components adopted an elastic-plastic constitutive model, and all tower structures were meshed with C3D8R solid elements. Tie constraints were applied to the blade-nacelle connection and flange interfaces. For modal analysis, the tower base was fully fixed. In seismic dynamic analysis, ground motion input was imposed at the base via acceleration amplitude

loading, and other directional degrees of freedom were constrained [36].

In the FE analysis, simulating physical joint damage by directly deleting bolt elements often introduces severe localized stress singularities and convergence issues. Accordingly, the elastic modulus reduction method based on the principle of equivalent bending stiffness is adopted to simulate node damage. Structural damage, such as concrete cracking, contact interface degradation, or connection looseness, inherently reduces the local bending stiffness EI of components. This localized deterioration can be macroscopically equivalently characterized by lowering the elastic modulus of the structure to reflect the stiffness degradation, which has been widely applied in structural damage numerical simulation [37–39]. Nevertheless, it should be acknowledged that reducing the elastic modulus is a simplified equivalent approach with certain limitations. It cannot capture the discrete distribution of local damage, nonlinear fracture behavior, crack propagation, or detailed energy dissipation mechanisms involved in actual damage evolution. Therefore, this method is suitable for global dynamic response analysis, rather than refined local damage simulation. Based on the principle of stiffness equivalence, the moment of inertia I_{damage} of the flange section after damage is calculated, then the equivalent elastic modulus E_{eq} after damage can be obtained by the following formula:

$$E_{\text{intact}} I_{\text{damage}} = E_{\text{eq}} I_{\text{intact}} \quad (1)$$

Where E_{intact} and I_{intact} represent the initial elastic modulus and moment of inertia of the intact flange, respectively. After calculation, the equivalent elastic modulus of the damaged flange is determined as $E_{\text{eq}} = 0.97 E_{\text{intact}}$. In addition, a friction contact with a coefficient of 0.4 was defined in the damaged flange area [40–42].

2.3. Analysis of test results

The acceleration envelope of the tower is shown in Fig. 3. It can be seen from the figure that under the action of the earthquake, the acceleration response of each damage condition model changes uniformly along the height of the tower. For the undamaged model, the peak acceleration occurs at the tower top, reaching 0.326 g with an acceleration amplification factor of 1.630. The maximum acceleration of the damage condition model appeared at the upper connecting flange of the model. Specifically, the maximum accelerations for the lower flange damage, upper flange damage, and multiple damage condition reach 0.384 g, 0.390 g, and 0.487 g, respectively. Compared with the maximum acceleration of the non damage condition, the maximum acceleration increased by 17.79 %, 19.63 % and 49.39 % respectively. The maximum acceleration amplification factors of each damage condition reached

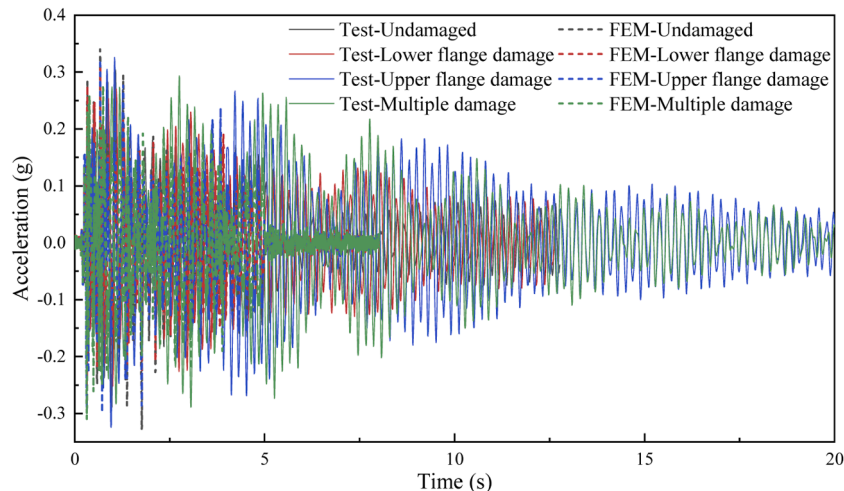


Fig. 4. Time history curve of tower top acceleration under different damage conditions.

1.920, 1.950 and 2.435, respectively. In the FE results, the maximum accelerations for the undamaged condition, the lower flange damaged condition, the upper flange damaged condition, and the multiple damage condition reached 0.341 g, 0.406 g, 0.361 g, and 0.450 g, respectively, with errors of 4.60 %, 5.73 %, 7.44 %, and 7.56 %, which are within a reasonable range of errors.

The acceleration time history curve of the tower under different damage conditions is shown in Fig. 4. By comparison, it can be seen that the finite element results fit well with the experimental results. The dynamic response of the damaged tower tube structure appears obvious beat vibration phenomenon after loading. That is, when two simple harmonic waves with different amplitudes but similar frequency are superimposed, the amplitude of their synthetic waveform changes

periodically with the vibration time. Set two response components:

$$\begin{aligned} X_1 &= A_1 e^{-\xi_1 \omega_1 t} \cos(\omega_1 t + \varphi_1) \\ X_2 &= A_2 e^{-\xi_2 \omega_2 t} \cos(\omega_2 t + \varphi_2) \end{aligned} \quad (2)$$

Where A_1, A_2 are the initial amplitudes of vibration in two orthogonal modal directions respectively; ξ_1, ξ_2 are two order orthogonal modal damping ratios respectively; ω_1, ω_2 are the circular frequencies of two order orthogonal modes respectively (the square term of damping ratio is ignored here), $\omega_1 = 2\pi f_1, \omega_2 = 2\pi f_2, f_1, f_2$ are two order orthogonal modal frequencies respectively; φ_1, φ_2 are the initial phases of the two vibrations respectively. The beat signal is actually the superposition of the above two signals, and there are:

$$X = X_1 + X_2 = \sqrt{\bar{A}_1^2 + \bar{A}_2^2 + 2\bar{A}_1\bar{A}_2\cos[(\omega_1 - \omega_2)t + \varphi_1 - \varphi_2]} \sin\left(\frac{\omega_1 + \omega_2}{2}t + \frac{\varphi_1 + \varphi_2}{2} + \Delta\varphi\right) \quad (3)$$

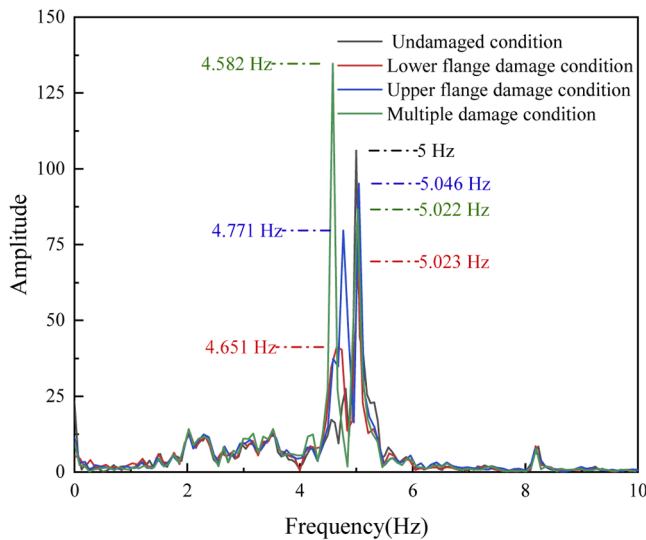


Fig. 5. Acceleration response power spectrum at the measuring point on the top of model tower.

Where $\bar{A}_1^2 = A_1^2 e^{-2\xi_1 \omega_1 t}, \bar{A}_2^2 = A_2^2 e^{-2\xi_2 \omega_2 t}$.

It can be seen that the beat signal can be seen as the product of the amplitude modulation term with $(\omega_1 - \omega_2)/2\pi$ frequency and the vibration signal with $(\omega_1 + \omega_2)/4\pi$ frequency (ignoring the influence of $\Delta\varphi$ periodic function on the frequency term).

The acceleration time history response is Fourier transformed to obtain the spectrum of the structure, as shown in Fig. 5. It can be found that under the undamaged condition, there is only one obvious peak in the Fourier spectrum value of the tower, and the first-order frequency of the tower is $f = 5.000$ Hz, the FE calculation result is $f = 5.5859$ Hz, with an error of 11.72 %. Under the damage condition of the tower, the Fourier spectrum value has two obvious peaks, corresponding to the first and second order frequencies of the tower. It can be considered that the existence of the beat vibration phenomenon leads to the simultaneous occurrence of the first-order front and rear shimmy frequency and the second-order left and right shimmy frequency of the model tower. The first and second order frequencies of the tower with lower flange damage are $f_1 = 4.651$ Hz and $f_2 = 5.023$ Hz, respectively (FE results are $f_1 = 5.1164$ Hz and $f_2 = 5.3519$ Hz, with errors of 10.01 % and 6.55 %, respectively). The first and second order frequencies of the tower with upper flange damage are $f_1 = 4.771$ Hz and $f_2 = 5.046$ Hz, respectively (FE results are $f_1 = 5.1566$ Hz and $f_2 = 5.3985$ Hz, with errors of 8.08 % and

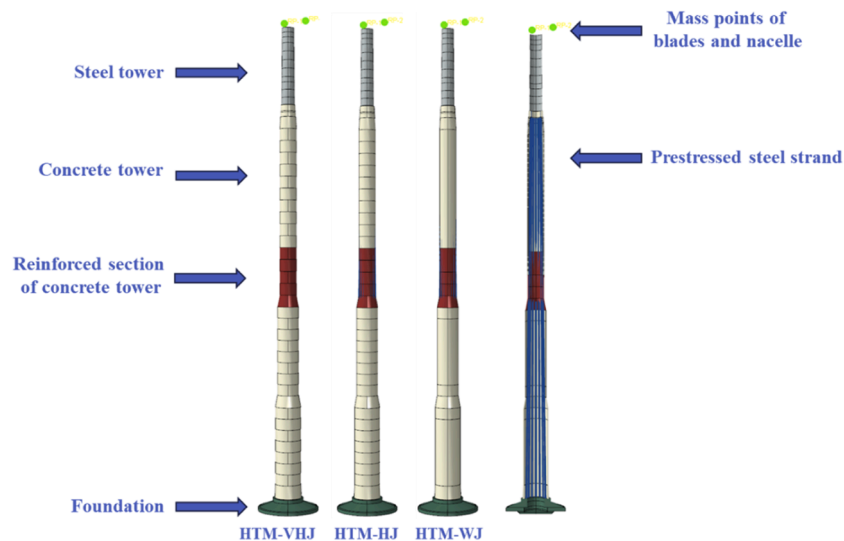


Fig. 6. FE model of the prestressed steel-concrete hybrid wind turbine tower.

Table 3

The parameters of the nacelle and the rotor.

Part	Mass (t)	Moment of inertia (kg.mm ²)
Nacelle	138.0	9.11×10^{10} , 4.34×10^{11} , 4.71×10^{11}
Rotor	117.1	5.43×10^{13} , 2.11×10^{13} , 2.22×10^{13}

Table 4

Material parameters of concrete.

Parameter	C80	C75	C45
Elastic modulus E_s (GPa)	38.0	37.5	33.5
Poisson's ratio ν		0.2	
Dilation angle ψ (°)		30	
Eccentricity e		0.1	
Ratio biaxial to uniaxial compressive strength f_{bo}/f_{co}		1.16	
Second stress invariant ratio K		2/3	
Viscosity parameter μ		0.0005	

Table 5

Maximum load of tower top in all directions.

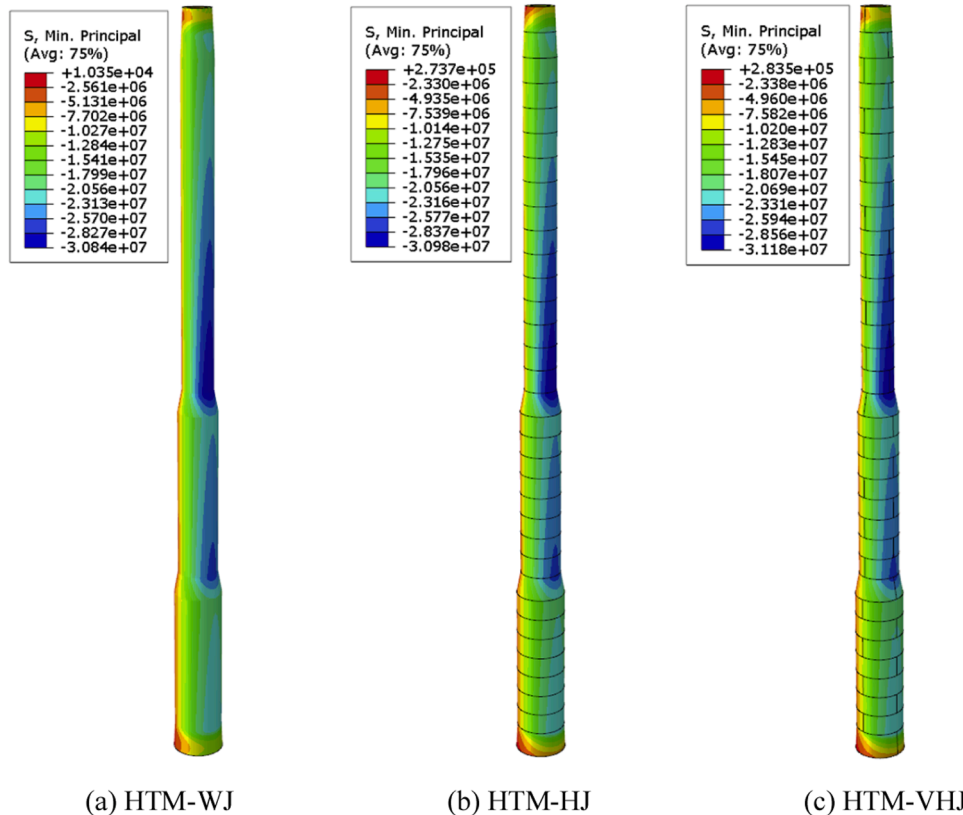
F_{xy} (kN)	F_z (kN)	M_{xy} (kN-m)	M_z (kN-m)
1092.66	-3613.14	15,773.00	12,894.90

6.99 %, respectively). And the first and second order frequencies of the tower with multiple flange damage are $f_1=4.582$ Hz and $f_2=5.022$ Hz, respectively (FE results are $f_1=5.0110$ Hz and $f_2=5.4458$ Hz, with errors of 9.36 % and 8.44 %, respectively). The FE errors are within a reasonable range. According to the above analysis, the beat wave period $T=1/|f_1-f_2|$ in the acceleration time history curve, the beat wave periods of the lower flange damaged tower model, the upper flange damaged tower model and the multi damaged tower model are

calculated to be 2.688 s, 3.636 s and 2.273 s respectively, which are basically consistent with the beat wave periods 2.585 s, 3.765 s and 2.190 s in the acceleration time history curve of each damage condition, further verifying the correctness of modal frequency identification. However, the rigidly fixed foundation used in this study amplifies nonlinear modal coupling, whereas realistic soil-structure interaction (SSI) introduces substantial radiation damping that could attenuate this beat-like phenomenon. In addition, practical structural health monitoring (SHM) typically relies on ambient operational wind loads rather than extreme seismic events. The detectability and applicability of such beat vibration in daily engineering practice require further validation.

3. FE model of the prestressed hybrid tower

Based on a prestressed hybrid tower structure from an actual engineering project, the FE model of the tower and foundation was established using ABAQUS software, as shown in Fig. 6. The tower has a total height of 157.633 m, comprising a 130.230 m tall concrete lower section and a 27.403 m tall steel upper section. The wall thickness of the concrete tower sections located at elevations of 5.07 m to 26.91 m and 34.47 m to 59.95 m is 0.285 m, the wall thickness of the concrete tower sections located at elevations of 67.51 m to 126.31 m is 0.305 m, and the wall thickness of the doorway segment is 0.64 m. The concrete tower has transition sections with a wall thickness of 0.33 m at bottom elevations of 30.55 m and 63.59 m, and a connecting section with a wall thickness of 0.75 m connecting to the top steel tower at a bottom elevation of 126.31 m. The prestress of the hybrid tower structure is applied through 16 externally prestressed steel strands that are evenly distributed in a circumferential direction. To simplify the simulation of the nacelle and blades, two mass points were defined based on the eccentric coordinates, moments of inertia, and actual weight distributions provided by the manufacturer. The parameters of the wind turbine nacelle and blades are shown in Table 3.

**Fig. 7.** Compressive stress contour of concrete tower with different joint schemes.

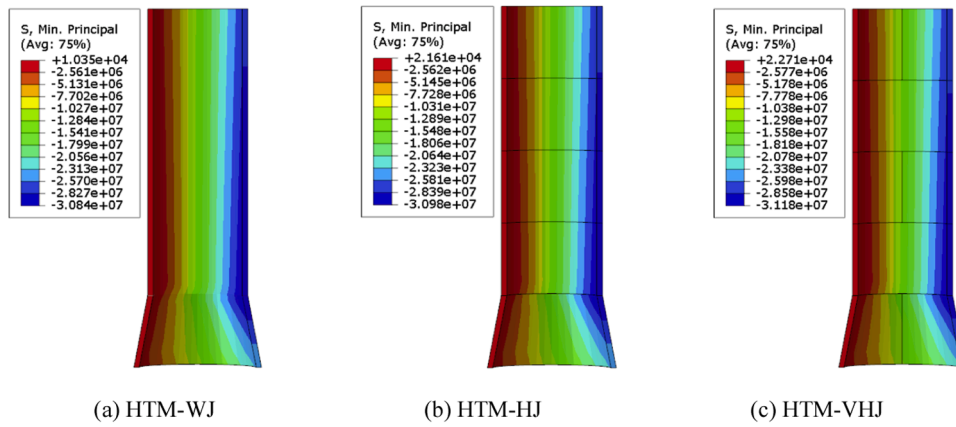


Fig. 8. Compressive stress contour of reinforced concrete tower sections with different joint schemes.

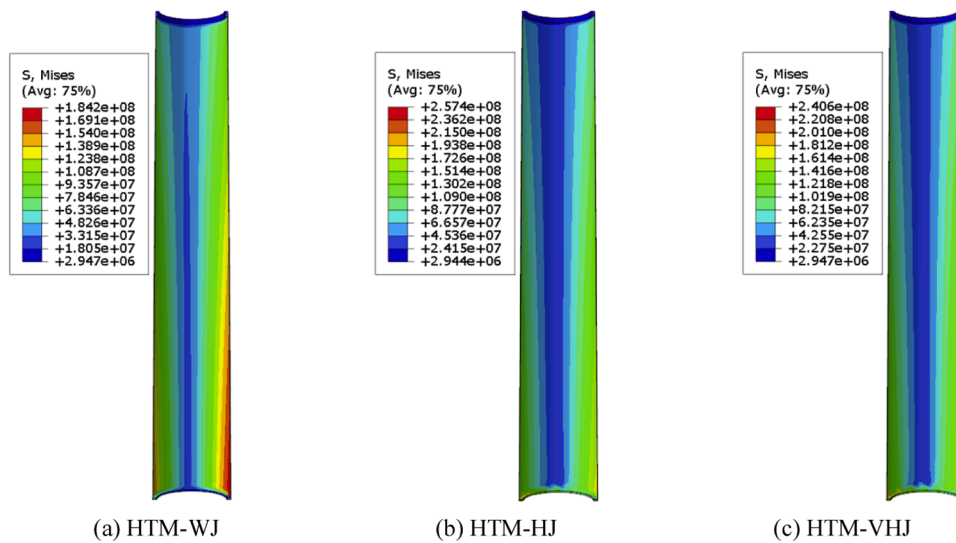


Fig. 9. Mises stress contour of steel tower with different joint schemes.

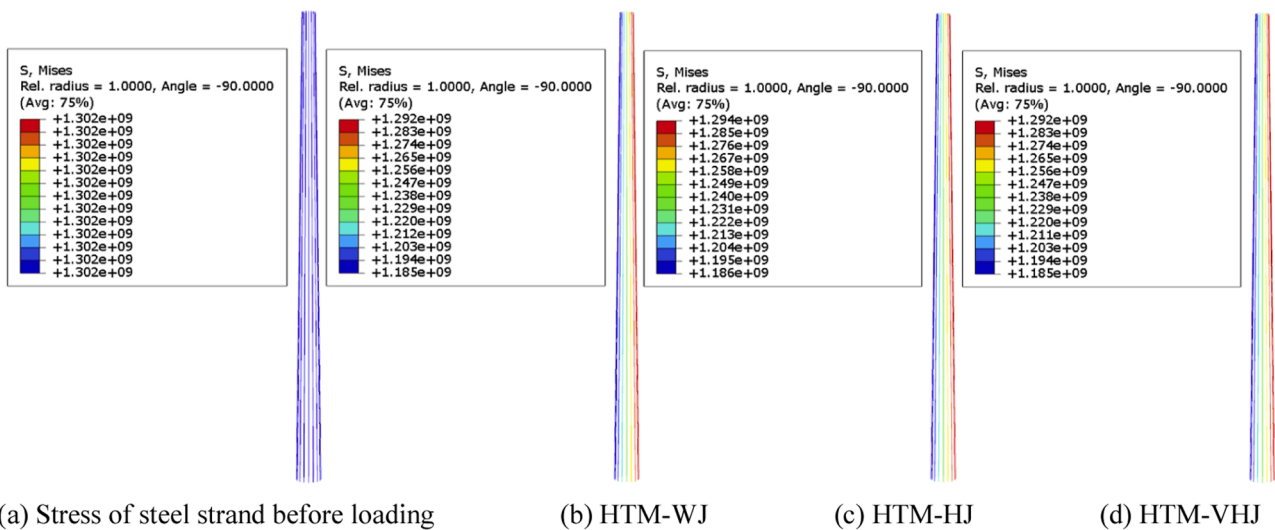


Fig. 10. Mises stress contour before and after loading of prestress steel strand with different joint schemes.

The lower concrete tower has 5 reinforced concrete tower sections at a height of 63.59 m to 83.19 m, with C80 concrete material. The

concrete material of the remaining concrete tower sections is C75, and the concrete material of the foundation is C40. The nonlinear behavior

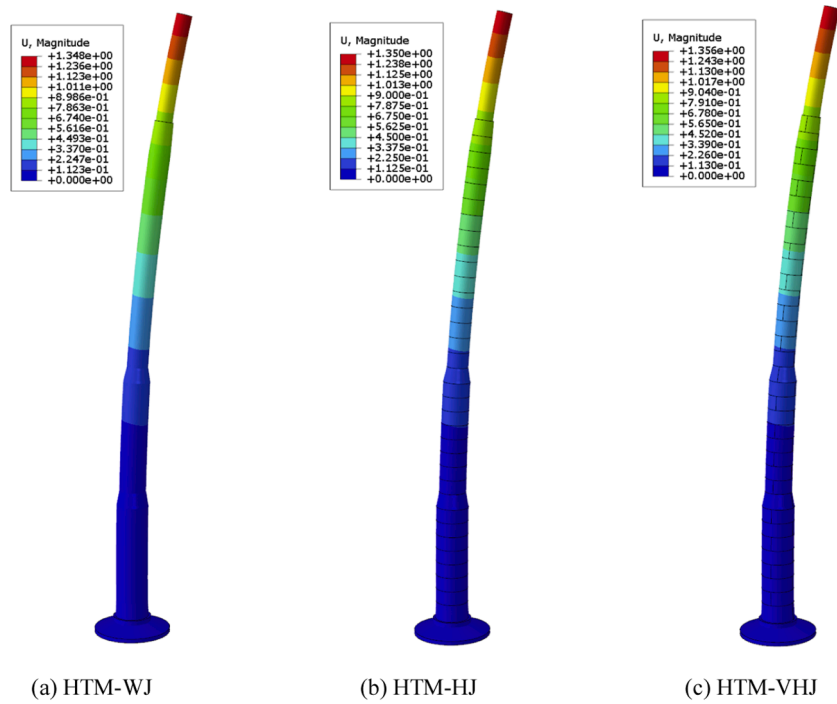


Fig. 11. Displacement contour of the prestressed hybrid tower with different joint schemes.

of the concrete is simulated using the Concrete Damaged Plasticity (CDP) model, and the material parameters are shown in Table 4. The mesh element type is C3D8R solid element. The upper steel tower is made of Q355 steel and is simulated using an elastoplastic model with S4R shell element mesh elements. The prestressed steel strands of the hybrid tower are 1860 MPa steel strands, and an elastoplastic model is used for simulation, with the mesh element type being B31 beam element. In the FE model, the wind turbine blades and the wind turbine nacelle, as well as the center of the upper surface of the wind turbine nacelle and the tower flange, are set as coupling. The tower and the foundation are rigidly connected by tie constraint, and the concrete tower and the steel tower are also constrained by tie contact. The top and bottom of the steel strands are constrained by tie contact with the top transition section of the concrete tower and the bottom of the foundation, respectively, and the prestress is applied by the decreasing temperature method. The bottom of the foundation is fully fixed, the three translational directions (U1, U2, U3) and the three rotational directions (UR1, UR2, UR3) are all set to zero, and all degrees of freedom are locked, simulating a fixed connection of the foundation.

To clarify the impact of the splice joints of the concrete tower on the modes and vibration modes of the overall prestressed hybrid tower structure, three sets of FE models were established: a hybrid tower model without joints (HTM-WJ), a hybrid tower model with horizontal joints (HTM-HJ), and a hybrid tower model with both vertical and horizontal joints (HTM-VHJ). In the HTM-HJ, the surface-to-surface contact is used to simulate the interaction of the horizontal joints of the concrete cylinder section. The normal behavior of the contact is defined by "hard contact" based on the classical Lagrange multiplier method, and the tangential behavior is defined by the penalty stiffness method. The friction coefficient between the contact surfaces is taken as 0.5. In the HTM-VHJ, each section is divided into two concrete segments. The vertical joints of adjacent concrete sections are staggered by 90°, and the vertical joints between the segments are set as Tie contact.

Table 5 shows the maximum loads in each direction at the turbine hub of the wind turbine generator set in this wind farm, provided by the wind turbine manufacturer. In the static analysis FE model, a coupling point is set at the height of the turbine hub, and the external load combination is applied to this coupling point in the direction of the load

combination according to the most unfavorable situation for the wind turbine generator set.

4. Analysis of the influence of joint schemes on the static and dynamic response of the hybrid tower structures

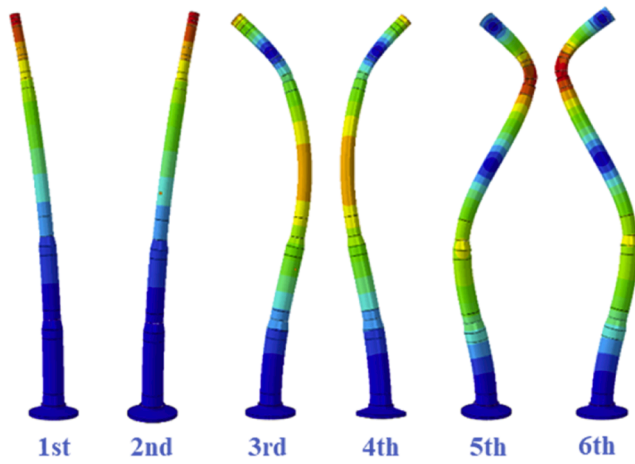
4.1. Static analysis of hybrid tower structures with different joint schemes

Fig. 7 shows the compressive stress contour of the hybrid tower under different joint schemes. It can be observed that under the combined action of external loads, the compressive stress concentration areas of the concrete tower are all distributed at the transition section of the variable cross-section in the loading direction, and the area of the compressive stress concentration area near the transition section at the central height of the tower is significantly larger than that near the transition section closer to the ground. The comparison shows that as the number of joints in the concrete tower increases, the maximum compressive stress of the structure also gradually increases. The maximum compressive stress of HTM-WJ, HTM-HJ, and HTM-VHJ reached 30.84 MPa, 30.98 MPa, and 31.18 MPa, respectively, all of which are less than the standard value of 44.50 MPa for the compressive strength of C75 concrete. Compared to HTM-WJ, the maximum compressive stress of HTM-HJ and HTM-VHJ increased by 0.14 MPa and 0.34 MPa, respectively, representing increases of 0.45 % and 1.10 %. This indicates that different joint schemes do not significantly affect the stress distribution of the concrete tower structure.

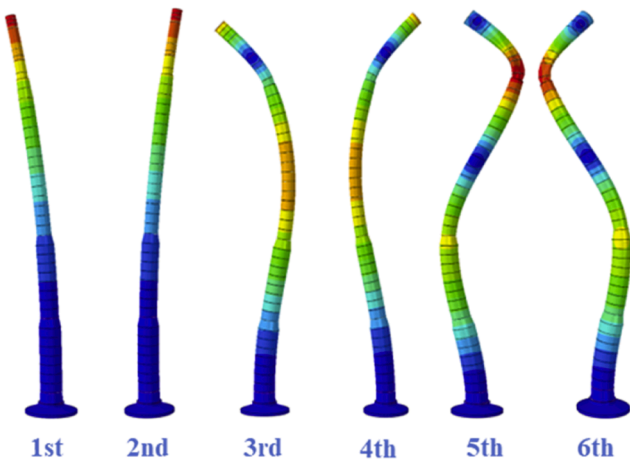
The reinforced composite tower section of the hybrid tower structure was subjected to local stress analysis, as shown in Fig. 8. The design strength grade of the tower is based on the characteristic cylindrical compressive strength f_{ck} determined over 28 days. The design value of the concrete compressive strength is defined according to the European standard for concrete structure design EN 1992-1-1: 2004 [43], that is

$$f_{cd} = \alpha_{cc}(f_{ck} / \gamma_c) \quad (4)$$

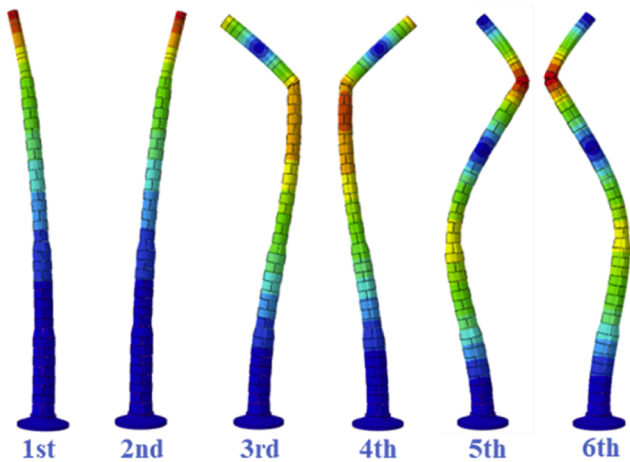
Where α_{cc} is a coefficient combining the long-term effect on compressive strength and the adverse effects caused by the load application method, $\alpha_{cc}=1.0$; f_{ck} is the characteristic of the compressive strength of the



(a) Mode shapes of each order of HTM-WJ



(b) Mode shapes of each order of HTM-HJ



(x) Mode shapes of each order of HTM-VHJ

Fig. 12. Mode shapes of the prestressed hybrid wind turbine tower under different joint states.

reinforced C80 concrete section within 28 days, $f_{ck}=50.2 \text{ N/mm}^2$; γ_c is the partial safety factor for concrete, $\gamma_c=1.5$. The calculated $f_{cd}=33.47 \text{ N/mm}^2$. It can be seen that the stress on the reinforced concrete tower section meets the specifications under different joint schemes.

Fig. 9 shows the Mises stress contour of the steel tower under

different joint schemes. It can be observed that under the combined action of external loads, the stress concentration areas of the steel tower are all distributed at the transition section of the variable cross-section in the loading direction. Through comparison, it can be found that the change in the number of joints of the concrete tower also affects the stress condition of the top steel tower. With the increase in the number of hybrid tower joints, the stress concentration area gradually decreases, and finally concentrates on the bottom flange of the connection between the steel tower and the concrete tower. The maximum Mises stresses of HTM-WJ, HTM-HJ, and HTM-VHJ reached 184.2 MPa, 217.0 MPa, and 240.6 MPa, respectively. Compared with the steel tower of HTM-WJ, the maximum Mises stresses of the steel towers of HTM-HJ and HTM-VHJ increased by 32.8 MPa and 56.2 MPa, respectively, with increases of 17.81 % and 30.62 %. This may be due to stress redistribution caused by changes in overall structural stiffness and the weakening of the overall continuity.

Fig. 10 shows the Mises stress contour of the prestress steel strands before and after loading under different joint schemes. Comparison reveals that, under combined external loads, the maximum prestress loss of the steel strands in HTM-WJ, HTM-HJ, and HTM-VHJ reached 116 MPa, 115 MPa, and 115 MPa, respectively, after loading. This indicates that different hybrid tower joint schemes have almost no impact on the structural prestress loss.

Fig. 11 shows the displacement contour of the prestressed hybrid tower under different joint schemes. By comparison, it can be found that the maximum displacements at the top of the tower for HTM-WJ, HTM-HJ, and HTM-VHJ reached 1.35 m, 1.35 m, and 1.36 m, respectively. This indicates that the change in the number of joints of the concrete tower has no significant impact on the deformation of the overall prestressed hybrid tower structure. According to the structural stiffness requirements, the maximum displacement $D_{\max}$ of the tower is $<1\%$ of the tower height H [44], that is

$$D_{\max} \leq \frac{H}{100} = 1.6m \quad (5)$$

It can be observed that the maximum displacement at the top of the hybrid tower meets the specifications under different joint schemes.

4.2. Dynamic analysis of hybrid tower structures with different joint schemes

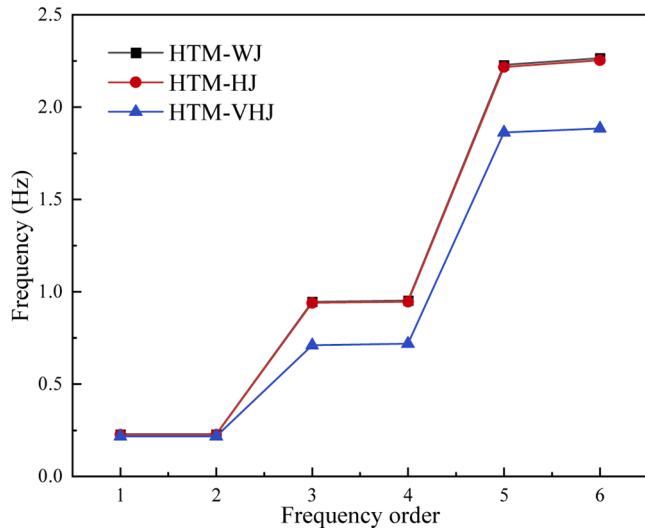
The mode shapes of the simplified tower models with three different joint schemes were calculated using ABAQUS FE software, as shown in Fig. 12. Since the steel-concrete hybrid tower structure is a symmetrical structure, and the first and second, third and fourth, and fifth and sixth order frequencies are basically equal in value, the vibration modes with similar frequencies can be regarded as the same mode of the hybrid tower structure deforming in different directions. The first and second order vibration modes are swing modes. Under these modes, the top of the tower tilts and oscillates along the X-axis and Y-axis, respectively. The third and fourth vibration modes are bending modes, under which the tower exhibits obvious bending deformation in the middle. The fifth and sixth modes are bending-torsion combination modes. Under these modes, the tower exhibits an S-shaped combined bending and torsional deformation. A comparative analysis reveals that while horizontal joints have a negligible impact on the global mode shapes, the introduction of vertical joints significantly alters the higher-order modal profiles. Under the third and fourth bending modes, the bending point of HTM-VHJ shifts significantly upward. Furthermore, under the fifth and sixth bending-torsion combination modes, the bending degree at the connection between the concrete tower section and the steel tower section is more severe. This indicates that the appearance of vertical joints impairs the stiffness of the hybrid tower structure, leading to a redistribution of modal energy.

The natural frequencies and vibration mode data of simplified models of different jointed towers are presented in Table 6 and Fig. 13.

Table 6

First six natural frequencies of prestressed steel-concrete hybrid wind turbine tower.

Frequency order	HTM-WJ		HTM-HJ		HTM-VHJ		Mode characteristics
	Frequency (Hz)	Mean frequency (Hz)	Frequency (Hz)	Mean frequency (Hz)	Frequency (Hz)	Mean frequency (Hz)	
1	0.2287	0.2287	0.2268	0.2269	0.2173	0.2174	Swing mode
2	0.2287		0.2269		0.2174		
3	0.9456	0.9490	0.9399	0.9433	0.7110	0.7152	Bending mode
4	0.9524		0.9466		0.7194		
5	2.2279	2.2464	2.2163	2.2346	1.8635	1.8740	Bending-torsion combination mode
6	2.2648		2.2529		1.8845		

**Fig. 13.** Variation of the first six frequencies of prestressed steel-concrete hybrid wind turbine tower.**Table 7**

Parameter analysis condition of prestressed steel-concrete hybrid wind turbine tower.

Case	Variation of elastic modulus	Changed elastic modulus (GPa)		Damage Variable D
		C75	C80	
HTM-HJ- $D = 0.05$	0.95 E_0	35.625	36.1	0.05
HTM-VHJ- $D = 0.05$				
HTM-HJ- $D = 0.10$	0.90 E_0	33.75	34.2	0.10
HTM-VHJ- $D = 0.10$				
HTM-HJ- $D = 0.15$	0.85 E_0	31.875	32.3	0.15
HTM-VHJ- $D = 0.15$				

Compared to HTM-WJ, when the tower body only has horizontal joints, the horizontal joints have a minimal impact on the natural frequencies of each order of the hybrid tower structure. Specifically, the presence of the horizontal joint has the greatest effect on reducing the structural frequency in the tower oscillation mode, from 0.2287 Hz to 0.2269 Hz, but the reduction is only 0.79 %. The simultaneous appearance of horizontal and vertical joints resulted in a significant decrease in the frequencies of each order of the tower. Compared with HTM-WJ, the reduction in the swing mode, bending mode, and bending-torsion

combined mode reached 4.94 %, 24.64 %, and 16.58 %, respectively. According to data provided by the wind turbine manufacturer, when the wind turbine speed is 9.29 rpm, the corresponding wind turbine blade rotation frequencies are 0.155 Hz for 1P, 0.465 Hz for 3P, and 0.93 Hz for 6P. Under these operational conditions, the harmonic frequency of the wind turbine generator is close to the second-order frequency of the tower. Furthermore, in actual engineering projects, the misalignment of concrete joints, horizontal joints, and vertical joints caused by construction reasons weaken the overall stiffness of the tower to a certain extent. The second-order frequency of the tower may be further reduced, which may lead to resonance due to overlap with the 6P rotational frequency of the wind turbine blades, and have a significant impact on the local strength and fatigue life of the structure. Consequently, the higher-order dynamic responses of hybrid towers demand rigorous evaluation during the structural design phase.

5. Analysis of the influence of joint damage on the static and dynamic response

To investigate the impact of different degrees of defects and damage on the dynamic characteristics of the prestressed hybrid tower structure, a series of dynamic response parameter analyses were conducted. The HTM-HJ and HTM-VHJ models were used for calculation, and the reduction in elastic modulus is employed to simulate structural damage based on fundamental damage mechanics. In SHM of wind turbine towers, identifying early-stage deterioration is the primary goal. Empirical studies show that a 5 % to 15 % stiffness loss typically corresponds to the onset of minor joint looseness or micro-cracking, providing a highly realistic range for dynamic sensitivity analysis and early fault detection [45–47]. Accordingly, the reduction coefficients of 0.95 E_0 , 0.90 E_0 and 0.85 E_0 (representing 5 %, 10 %, and 15 % reductions) were selected to simulate early-to-moderate structural damage. Joint damage was quantified by the damage variable D defined as $D = 1 - E/E_0$, where E and E_0 are the elastic modulus of concrete after damage and the elastic modulus of concrete before damage, respectively. The calculation conditions are shown in Table 7.

5.1. Static analysis of hybrid tower structures with different joint damage

Fig. 14 shows the compressive stress contour of the concrete tower under different degrees of joint damage, the spatial distribution of compressive stress concentrations remains largely unaffected by the extent of joint damage. The compressive stress concentration areas are all distributed at the transition sections of the variable cross-section in the loading direction, and the area of the compressive stress concentration area near the transition section at the central height of the tower is significantly larger than that near the transition section closer to the ground. The comparison reveals that as the degree of damage to the concrete tower joints increases, the maximum compressive stress of the hybrid tower structure with different joint schemes also gradually decreases. When the hybrid tower only has horizontal joints, the maximum compressive stress of HTM-HJ- $D = 0.05$, HTM-HJ- $D = 0.10$, and HTM-

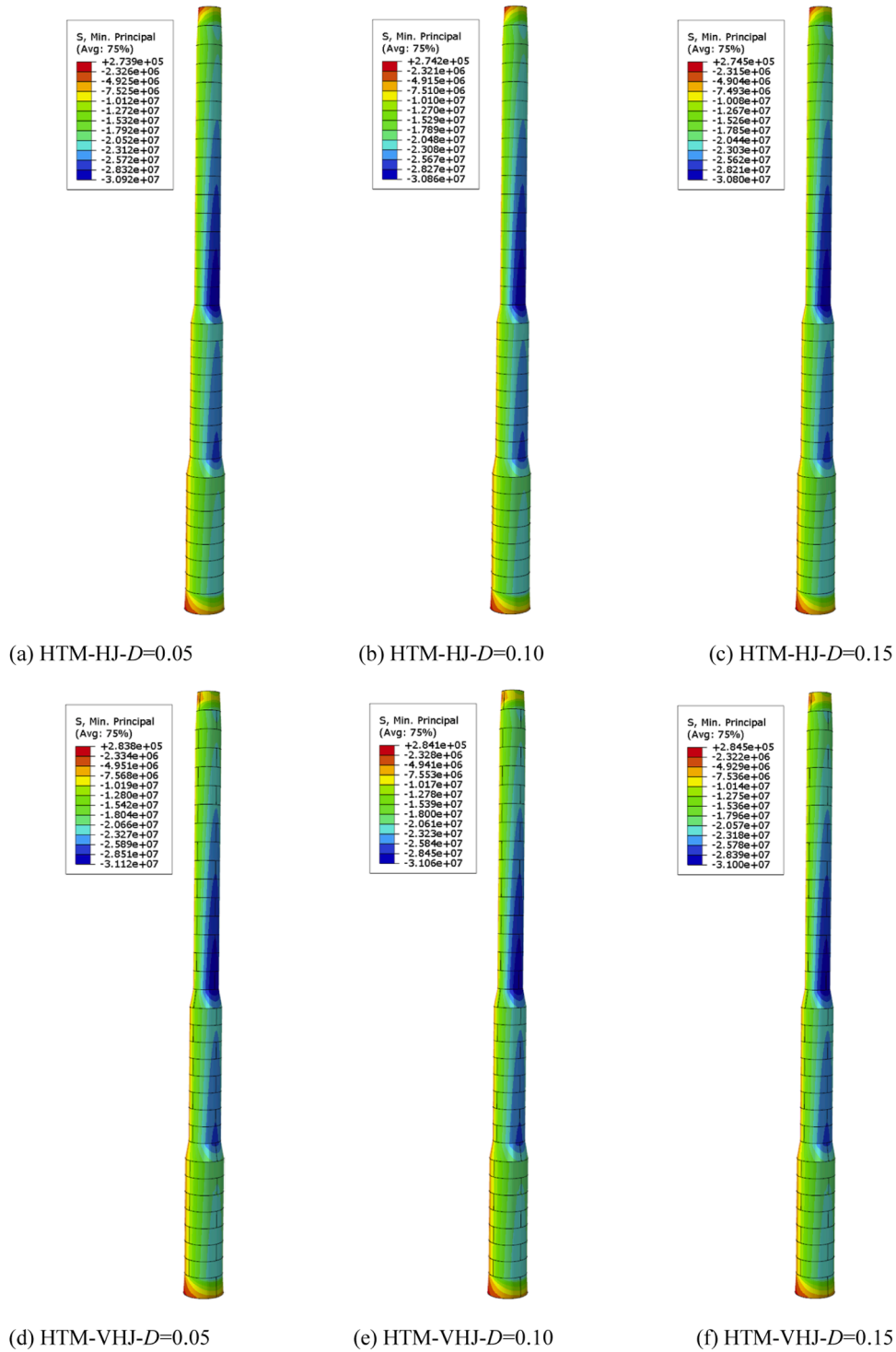


Fig. 14. Compressive stress contour of concrete tower under different degrees of joint damage.

HJ- $D = 0.15$ reached 30.92 MPa, 30.86 MPa, and 30.80 MPa, respectively, all of which are less than the standard value of compressive strength of C75 concrete of 44.50 MPa. Compared with the maximum compressive stress of 30.98 MPa of the hybrid tower structure when the joints are undamaged, the maximum compressive stress of HTM-HJ- $D = 0.05$, HTM-HJ- $D = 0.10$, and HTM-HJ- $D = 0.15$ decreased by 0.19 %, 0.39 %, and 0.58 %, respectively. When the hybrid tower has vertical and horizontal joints, the maximum compressive stresses of HTM-VHJ- $D = 0.05$, HTM-VHJ- $D = 0.10$, and HTM-VHJ- $D = 0.15$ reach 31.12 MPa, 31.06 MPa, and 31.00 MPa, respectively, which are all less than the

standard value of 44.50 MPa for the compressive strength of C75 concrete. Compared with the maximum compressive stress of 31.18 MPa in the hybrid tower structure when the joints are undamaged, the maximum compressive stress of HTM-VHJ- $D = 0.05$, HTM-VHJ- $D = 0.10$, and HTM-VHJ- $D = 0.15$ decreased by 0.19 %, 0.38 %, and 0.58 %, respectively. A cross-comparison indicates that the introduction of vertical joints induces a consistent, yet minor, stress increment of 0.20 MPa, with an increase of 0.65 %, across all equivalent damage states. Ultimately, this demonstrates that the stress sensitivity to progressive joint damage remains remarkably uniform, regardless of the specific joint

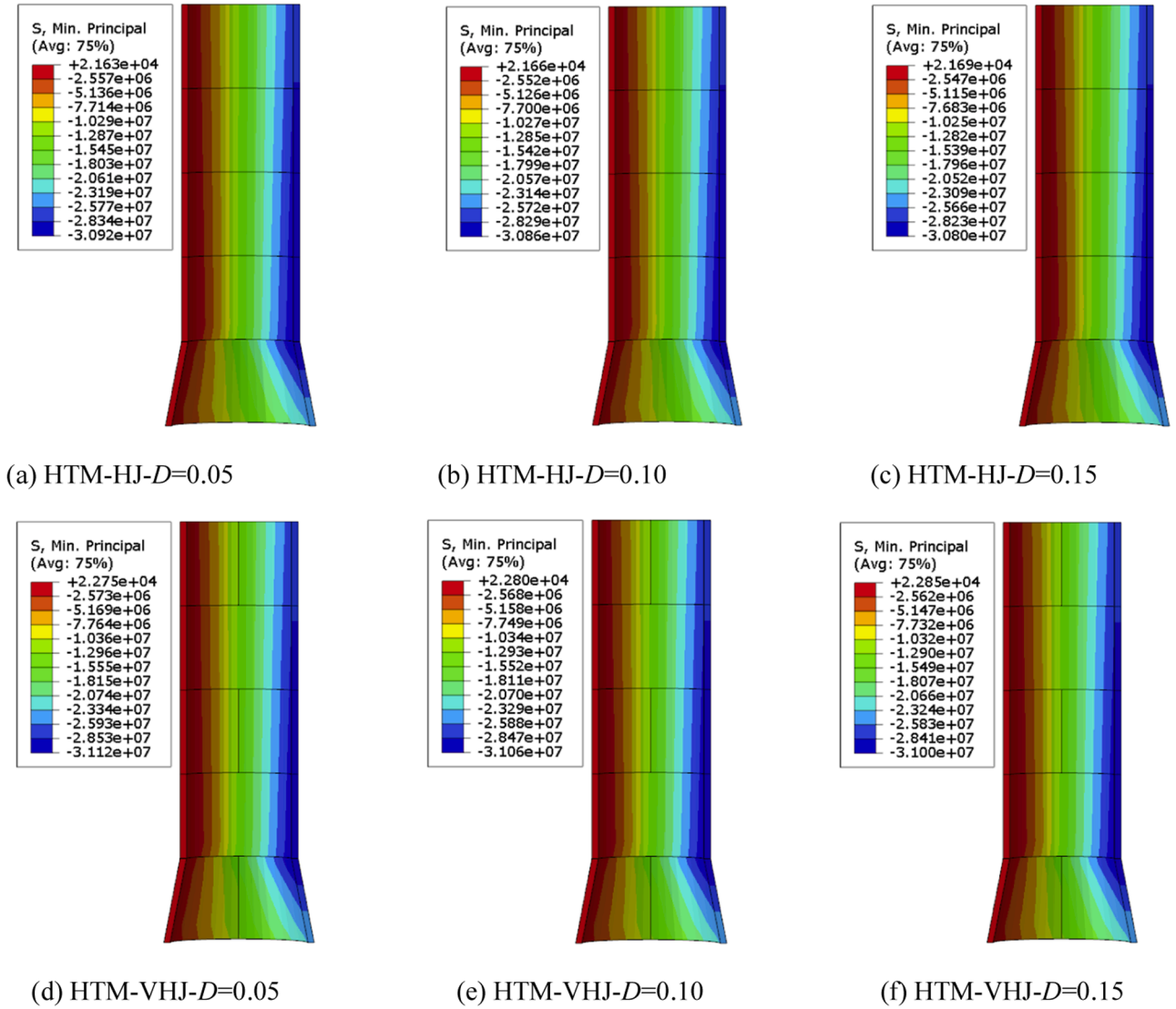


Fig. 15. Compressive stress contour of reinforced concrete tower sections with different degrees of joint damage.

configuration.

Fig. 15 shows the compressive stress contour of the reinforced hybrid tower section. It can be observed that, under different degrees of damage, the requirement of less than $f_{cd}=33.47 \text{ N/mm}^2$ is met. By comparing the stress distribution of the reinforced hybrid tower section under different degrees of joint damage, it can be found that joint damage does not have a significant impact on the stress distribution of the structure. The stress distribution at the joint is relatively uniform, and the degree of change in the maximum compressive stress of the structure is almost the same under different joint schemes.

Fig. 16 shows the displacement contours of the prestressed hybrid tower under different degrees of joint damage. Under combined external loading, as the degree of damage to the joints of the concrete tower increases, the maximum displacement at the top of the hybrid tower structure with different joint schemes also gradually increases. The maximum displacements at the top of the towers HTM-HJ- $D = 0.05$, HTM-HJ- $D = 0.10$, and HTM-HJ- $D = 0.15$ reached 1.42 m, 1.50 m, and 1.58 m, respectively. It can be observed that the maximum displacement at the top of the hybrid tower meets the specifications under different degrees of joint damage. Compared with the maximum displacement of 1.35 m at the top of the hybrid tower structure with only horizontal joints and no joint damage, the increases were 5.19 %, 11.11 %, and 17.04 %, respectively. The maximum displacements at the top of the

prestressed hybrid tower for HTM-VHJ- $D = 0.05$, HTM-VHJ- $D = 0.10$, and HTM-VHJ- $D = 0.15$ reached 1.43 m, 1.50 m, and 1.59 m, respectively. It can be observed that the maximum displacement at the top of the hybrid tower meets the specifications under different degrees of joint damage. Compared with the maximum displacement of 1.36 m at the top of the hybrid tower structure with vertical and horizontal joints when the joints are undamaged, the increases were 5.15 %, 10.29 %, and 16.91 %, respectively. A cross-comparison reveals that introducing vertical joints only marginally amplifies the top displacement by a maximum of 0.01 m, with an increase of 0.70 %, under equivalent damage states. This indicates that when the joints of the concrete tower sections in a hybrid tower structure are damaged to varying degrees, the degree of impact on the deformation of the concrete tower sections with different joint schemes is almost the same, and all of them meet the requirement that the maximum displacement $D_{\max}$ of the tower is $<1 \%$ of the tower height H .

5.2. Dynamic analysis of hybrid tower structures with different joint damage

Comparing the modal results of the most representative $D = 0.15$ model, it can be found that different damage states do not significantly change the modal shapes of the tower with horizontal joints and the

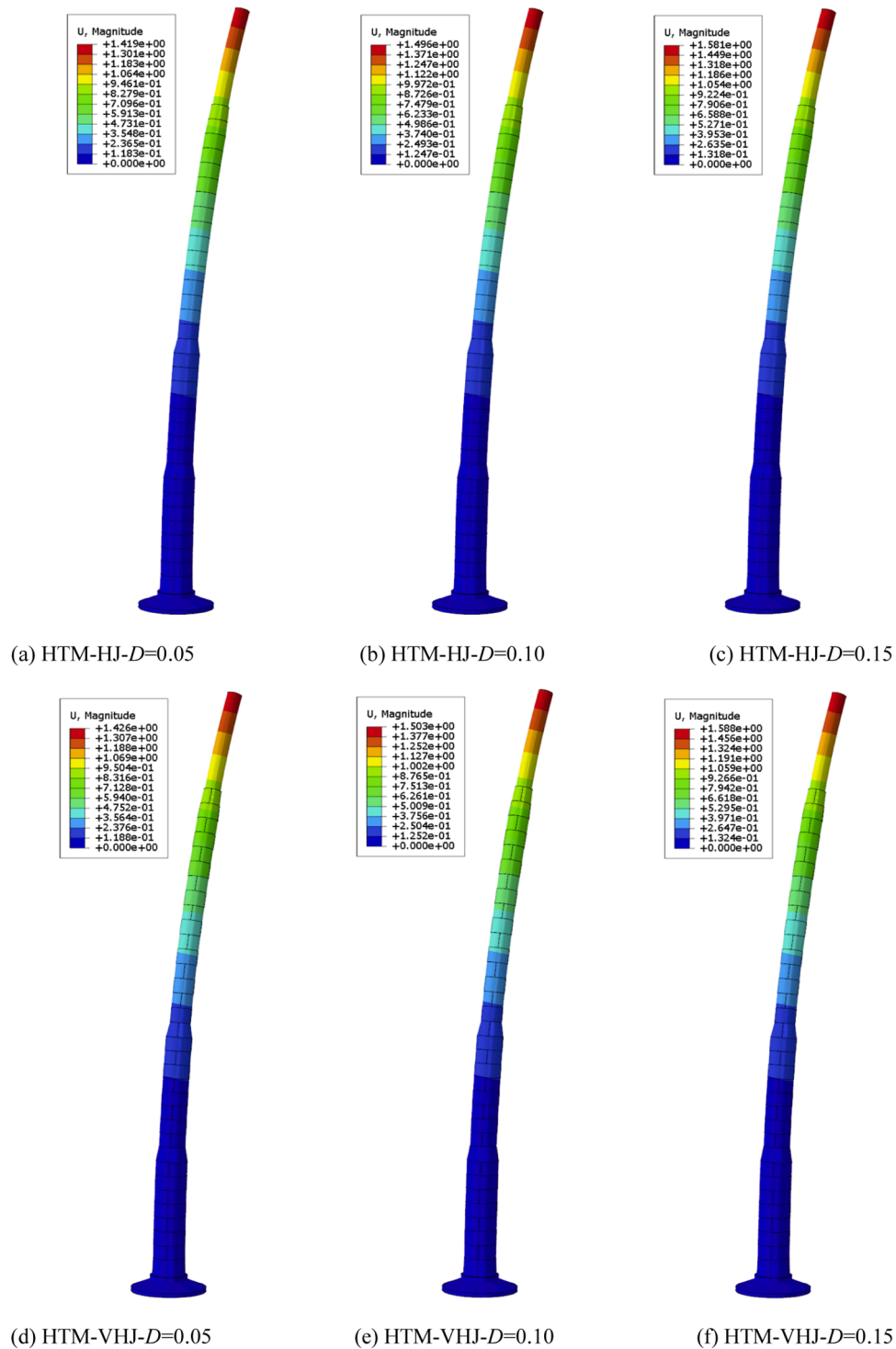


Fig. 16. Displacement contour of the prestressed hybrid tower under different degrees of joint damage.

tower with both horizontal and vertical joints. The bending points of each modal shape are almost identical to those of the original model, as shown in Fig. 17.

The mean frequencies under the same mode shape were compared and analyzed, and the results are summarized in Table 8. Comparing the FE results reveals that changes in the overall structural stiffness have a significant impact on the fundamental frequency of the hybrid tower. In the FE calculation results of the HTM-HJ, compared with the original model where the stiffness remained unchanged, when the damage variable D reaches 0.15, the average frequency of the first three mode

shapes of the structure decreased by 7.801 %, 7.802 %, and 7.805 %, respectively. A nearly identical trend is observed for the HTM-VHJ, where the damage variable $D = 0.15$ yields corresponding frequency reductions of 7.820 %, 7.802 %, and 7.801 %. By comparison, it can be found that the change in the stiffness of the overall structure has an almost consistent effect on the frequencies of each order of the hybrid tower, and the amplitude of change of each order of frequencies is also basically the same under different joint conditions.

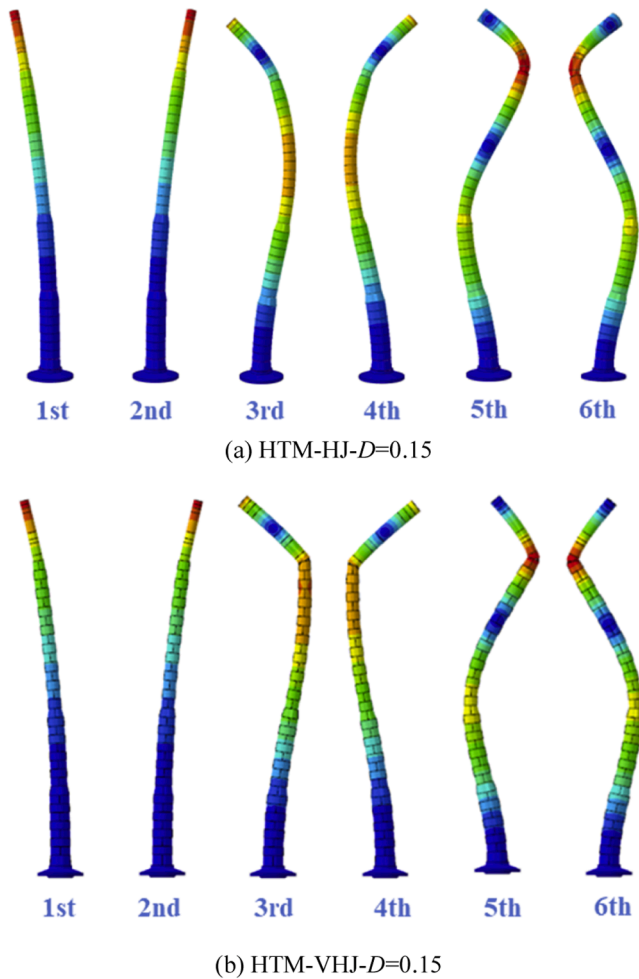


Fig. 17. Mode shapes of each order of the prestressed hybrid wind turbine tower.

Table 8

Frequency variation of prestressed steel-concrete hybrid wind turbine tower under different cases.

Mode shapes, Damage variable D	Frequency (Hz)					
	HTM-HJ			HTM-VHJ		
	Swing	Bending	Bending-torsion	Swing	Bending	Bending-torsion
0.00	0.2269	0.9433	2.2346	0.2174	0.7152	1.8740
0.05	0.2211	0.9195	2.1781	0.2119	0.6971	1.8266
0.10	0.2152	0.8949	2.1200	0.2062	0.6785	1.7779
0.15	0.2092	0.8697	2.0602	0.2004	0.6594	1.7278

6. Conclusion

In this study, a series of factors affecting the static and dynamic characteristics of the prefabricated prestressed steel-concrete hybrid tower structure are analyzed by using the method of shaking table test and numerical simulation. The main conclusions are as follows:

- (1) The shaking table tests successfully verified the FEM and revealed beat-like vibrations under specific seismic excitations. However, considering the current lack of numerical confirmation and ambient wind-load data, the detectability and applicability of such beat vibration in daily engineering practice require further validation.

- (2) The hybrid tower exhibits high static safety in the concrete section, but vertical joints cause significant stress concentration at the transition interface. Specifically, the maximum concrete displacement and compressive stress remain well within design limits. However, the introduction of vertical joints increases the peak Mises stress in the steel tower by 30.62 %, concentrating heavily at the bottom flange connection.
- (3) Vertical joints significantly weaken the higher-order dynamic stiffness of the structure, posing a critical resonance risk during operation. While horizontal joints cause a negligible frequency reduction of only 0.79 %, vertical joints drastically reduce the higher-order bending mode frequency by 24.64 %. In practice, this degradation pushes the structure's frequency dangerously close to the turbine's 6P rotational frequency, making rigorous high-frequency fatigue evaluation imperative during the design phase.
- (4) The concrete tower maintains high static safety under early-to-moderate structural damage. At a maximum damage variable of 0.15 representing a 15 percent structural degradation, the peak top displacement increases to 1.59 m, safely satisfying the strict 1 percent height limit. Simultaneously, the maximum compressive stress relaxes slightly to 31.00 MPa, remaining well below the 44.50 MPa capacity limit of the concrete material.
- (5) Early-to-moderate joint damage severely threatens the dynamic stability of the operating tower. Reaching a damage variable of 0.15 induces a uniform frequency reduction of approximately 7.8 percent across the fundamental modes.

CRedit authorship contribution statement

Xintong Hao: Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Huageng Hao:** Writing – original draft, Supervision, Resources, Investigation, Conceptualization. **Liying Zhang:** Writing – review & editing, Resources, Investigation, Funding acquisition. **Ying Xu:** Writing – original draft, Supervision, Resources, Investigation, Conceptualization. **Tianlu Li:** Resources, Investigation. **Yinting Yang:** Resources, Investigation. **Xiaohui Wang:** Resources, Investigation. **Haijun Wang:** Resources, Investigation. **Yaohua Guo:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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